

Xiaolong Tu

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AMAI Lab | 

EDUCATION

Ph.D. Candidate in Computer ScienceJan. 2023 – May 2027 (expected)

Georgia State University
Advisor: Dr. Haoxin Wang, Assistant Professor, Department of Computer Science

RESEARCH INTERESTS

Multimodal Foundation Models | LLM Training and Inference Efficiency | Model Benchmarking, Compression, and Neural Architecture Search

TECHNICAL SKILLS

- Programming:** Python, C/C++, Bash, Linux, Git
- Deep Learning:** PyTorch, Tensorflow, Hugging Face Transformers, vision-language models, Mamba and state-space models, GPU training, fine-tuning, distillation
- Efficiency & Evaluation:** Quantization, neural architecture search, latency and energy profiling, benchmark and dataset construction, end-to-end ML pipelines, llama.cpp / ONNX Runtime

RESEARCH EXPERIENCE

Research AssistantGeorgia State University, Atlanta, GeorgiaJan. 2023 – Present

- Efficient Hybrid Multimodal Models** *(Ongoing Research)*
 - Study the efficiency-accuracy trade-off in hybrid attention-state-space (SSM) multimodal architectures, and localize downstream performance degradation to parameter update behavior inside the state-space components rather than treating it as an aggregate score drop.
 - Analyze what is actually lost during distillation, tracing how representations and cross-modal alignment diverge between teacher and student to identify which signals fail to transfer.
 - Propose architectural and training interventions targeting the identified failure mode, restoring downstream performance while retaining the efficiency gains.
- LLM Efficiency and Evaluation**  **LM-Meter**
 - Built a fine-grained inference profiler for language models that exposes per-stage runtime behavior across mobile, embedded, and GPU platforms at minimal overhead.
 - Characterized prefill and autoregressive decoding latency, throughput, and memory pressure, isolating which architectural and serving choices dominate end-to-end cost.
 - Showed that prompt length, structure, and keyword composition measurably shift LLM latency and energy, establishing an empirical basis for cost-aware prompting.
- Model Search and Benchmarking Platforms**  **PlatformX**
 - Developed end-to-end platforms integrating hardware profiling, learned performance prediction, model search, and real-device validation in a single measurement-in-the-loop pipeline.
 - Designed transferable kernel-level cost predictors with few-shot calibration, generalizing to unseen hardware from a handful of measurements and cutting search cost substantially.
 - Released open benchmarks and datasets (DeepEn2023, AIEnergy) for studying model performance and energy consumption on real hardware.

INDUSTRY EXPERIENCE

Cellular Interoperability System EngineerApple China R&D, Beijing, ChinaOct. 2021 – Dec. 2022

- Designed and validated end-to-end 4G/5G wireless systems across mobile devices, radio access networks, and core networks.

- Built measurement and analysis workflows to diagnose interoperability and performance regressions in laboratory and commercial environments.

Modem Integration Engineer Qualcomm China R&D, Shanghai, China May 2021 – Oct. 2021

- Integrated and validated LTE modem platforms through protocol, interoperability, and performance testing.

Wireless System Interoperability Engineer Huawei Shanghai R&D, Shanghai, China Aug. 2018 – May 2021

- Led 5G NR interoperability verification and proof-of-concept system integration with device, chipset, and network partners.

Wireless Network Engineer Ericsson China R&D, Nanjing, China Jul. 2012 – Aug. 2018

- Designed and optimized large-scale 3G/4G radio access networks using field measurements and network performance analysis.

PUBLICATIONS

Papers Published or Accepted

[J2] X-EDGE: A Measurement-in-the-Loop Platform for Target-Aware Edge Multimedia Model Search

- Xiaolong Tu, Haoxin Wang, Dawei Chen, Yitao Chen, Kyungtae Han, and Jiang Xie

• *IEEE Transactions on Multimedia, Ready to Submit*

[C9] Keyword Matters: Unveiling the Energy Sensitivity of On-Device LLM Prompting

- Roy Tao, Xiaolong Tu, and Haoxin Wang

• *SEC Edge Intelligence Workshop 2026*

[C8] Transferable Latency Prediction for Fast LLM Screening on Heterogeneous Edge Devices

- Xiaolong Tu, Vinod K. Mishra, Venkat R. Dasari, Anu G. Bourgeois, and Haoxin Wang

• *ImmerCom 2026*

[C7] How Do Prompt Variations Affect Energy Consumption in On-Device LLMs?

- *Wei Hu, *Xiaolong Tu, Dawei Chen, Kyungtae Han, Onur Altintas, and Haoxin Wang

• *EMNLP, 2026, main conference* | [Project](#)

[C6] LM-Meter: Unveiling Runtime Inference Latency for On-Device Language Models. (**Best Paper Finalist**)

- Haoxin Wang, Xiaolong Tu, Hongyu Ke, Huirong Chai, Dawei Chen, and Kyungtae Han

• *ACM/IEEE Symposium on Edge Computing (SEC), Washington, D.C., 2025* | [Code](#)

[C5] PLATFORMX: An End-to-End Transferable Platform for Energy-Efficient Neural Architecture Search

- Xiaolong Tu, Dawei Chen, Kyungtae Han, Onur Altintas, and Haoxin Wang

• *ACM/IEEE Symposium on Edge Computing (SEC), Washington, D.C., 2025* | [Code](#)

[J1] AIEnergy: An Energy Benchmark for AI-Empowered Mobile and IoT Devices

- Xiaolong Tu, Anik Mallik, Haoxin Wang, and Jiang Xie

• *ITU Journal on Future and Evolving Technologies, Volume 6, Issue 2, 2025* | [Paper](#)

[C4] GreenAuto: An Automated Platform for Sustainable AI Model Design on Edge Devices

- Xiaolong Tu, Dawei Chen, Kyungtae Han, Onur Altintas, and Haoxin Wang

• *The 26th International Workshop on Mobile Computing Systems and Applications (HotMobile), Palm Springs, CA, 2025* | [Code](#) | [Paper](#)

[C3] Unveiling Energy Efficiency in Deep Learning: Measurement, Prediction, and Scoring across Edge Devices

- Xiaolong Tu, Anik Mallik, Dawei Chen, Kyungtae Han, Onur Altintas, Haoxin Wang, and Jiang Xie

• *ACM/IEEE Symposium on Edge Computing (SEC), Wilmington, DE, 2023* | [Project](#) | [Code](#)

[C2] DeepEn2023: Energy Datasets for Edge Artificial Intelligence

- Xiaolong Tu, Anik Mallik, Haoxin Wang, and Jiang Xie

• *NeurIPS 2023 Workshop: Tackling Climate Change with Machine Learning, New Orleans, Louisiana, 2023*

[C1] Visualization of Mobility Digital Twin: Framework Design, Case Study, and Future Challenges

- Yueyang Liu, Xiaolong Tu, Dawei Chen, Kyungtae Han, Onur Altintas, Haoxin Wang, and Jiang Xie

• *The 20th IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS), 2023*

PROFESSIONAL ACTIVITIES

Reviewer, IEEE Internet of Things Journal	Oct. 2023 – Present
Student Member, Institute of Electrical and Electronics Engineers (IEEE)	Mar. 2023 – Present
Student Member, Association for Computing Machinery (ACM)	Mar. 2025 – Present

ACADEMIC HONORS

The Martin D. Fraser Graduate Student Conference Travel Award (GSU)	2025
NSF Student Travel Grant Award and HotMobile Student Travel Award	2025
NSF Student Travel Grant Award and SEC Student Travel Award	2023
The Inaugural Martin D. Fraser Graduate Student Conference Travel Award (GSU)	2023